

AI RESPONSE QUALITY EVALUATOR AGENT

Design Document

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PROJECT: AI RESPONSE QUALITY EVALUATOR AGENT

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1. Executive Summary

Artificial Intelligence has transformed the way users interact with technology. Large Language Models (LLMs) such as ChatGPT, Gemini, Claude, and Llama are capable of answering questions, generating content, and assisting users in various domains. However, despite their impressive capabilities, these models may sometimes produce inaccurate or misleading information, commonly known as hallucinations.

The AI Response Quality Evaluator Agent is designed to automatically assess the quality of AI-generated responses. The proposed system evaluates responses using multiple quality dimensions, including factual accuracy, relevance, completeness, and hallucination detection. Instead of relying solely on the internal knowledge of the language model, the system uses a Retrieval-Augmented Generation (RAG) pipeline to retrieve trusted information from external knowledge sources.

The evaluation process is performed by multiple specialized agents. Each agent focuses on one quality metric and independently analyzes the response. Their individual scores are then combined by a Verdict Agent to produce an overall evaluation score and detailed feedback.

The system is designed using a modular architecture, making it scalable and easy to extend with additional evaluation metrics or support for multiple language models in future versions.

This document presents the complete design of the AI Response Quality Evaluator Agent, including the system architecture, evaluation workflow, data models, technology stack, and implementation approach for Milestone 1.

2. Introduction

Artificial Intelligence has become one of the most influential technologies in modern software development. Large Language Models (LLMs) are now widely used in education, healthcare, finance, customer support, software engineering, and research because they can generate human-like responses to natural language questions.

Although these models produce fluent and meaningful responses, they do not always guarantee factual correctness. Since they generate text based on learned patterns instead of performing real-time fact verification, they may sometimes provide inaccurate or unsupported information. These incorrect responses are referred to as hallucinations.

As organizations increasingly adopt AI-powered systems, ensuring the reliability and trustworthiness of AI-generated responses has become essential. Manual evaluation of AI responses is time-consuming and subjective. Therefore, there is a need for an automated evaluation system capable of measuring different aspects of response quality.

The AI Response Quality Evaluator Agent addresses this challenge by combining Retrieval-Augmented Generation (RAG) with a multi-agent evaluation framework. The system retrieves relevant reference information from trusted datasets and uses specialized evaluation agents to analyze different quality dimensions. This approach improves the reliability of the evaluation process and helps users understand the strengths and weaknesses of AI-generated responses.

The proposed system is intended to serve as the foundation for future intelligent evaluation platforms capable of supporting multiple language models, evaluation frameworks, and application domain

3. Problem Statement

Large Language Models generate responses based on statistical patterns learned during training rather than verifying facts from reliable knowledge sources. As a result, AI-generated responses may contain factual inaccuracies, incomplete information, or unsupported claims.

This limitation creates significant challenges in domains where accurate information is critical, such as healthcare, finance, education, and legal services. Users often have no reliable way to determine whether the generated response is trustworthy.

Existing evaluation methods are either manual, time-consuming, or limited to specific datasets. There is a growing need for an intelligent evaluation system that can automatically verify AI-generated responses using trusted reference information and provide quantitative quality scores.

The proposed AI Response Quality Evaluator Agent addresses this problem by integrating Retrieval-Augmented Generation (RAG) with multiple evaluation agents. The system compares AI-generated responses against verified knowledge, detects hallucinations, and provides an overall assessment based on multiple quality dimensions.

4. Objectives

The primary objectives of this project are:

- Design a modular AI response evaluation system.
- Evaluate the factual accuracy of AI-generated responses.
- Detect hallucinated or unsupported information.
- Measure response relevance with respect to the user question.
- Evaluate response completeness.
- Generate independent quality scores for each evaluation dimension.
- Produce an overall evaluation score using multiple agent outputs.
- Retrieve trusted knowledge through a Retrieval-Augmented Generation (RAG) pipeline.
- Build a reusable architecture for future AI evaluation systems.
- Improve the reliability and trustworthiness of AI-generated responses.

5. Scope of the Project

The scope of this project focuses on designing and developing the foundational architecture for an AI Response Quality Evaluator Agent as part of Milestone 1. The system accepts a user question, an AI-generated response, and an optional reference answer or source document through a unified evaluation interface.

The system retrieves supporting information from trusted datasets using a Retrieval-Augmented Generation (RAG) pipeline. Multiple evaluation agents independently analyze the AI response across several quality dimensions, including accuracy, relevance, completeness, and hallucination detection. Their results are aggregated to generate a final evaluation report.

The current milestone emphasizes system design, architecture, data models, evaluation workflow, and knowledge base creation. Full-scale implementation, advanced dashboards, and optimization of evaluation models are considered future enhancements beyond Milestone 1

6. Literature Study

Before designing an AI response evaluation system, it is essential to understand the key concepts that influence the quality and reliability of AI-generated responses. This section explains the major technologies, methods, and evaluation frameworks used in the proposed system.

The study focuses on four important areas:

- Large Language Model (LLM) Evaluation Techniques
- Hallucination Detection
- Retrieval-Augmented Generation (RAG)
- Existing Evaluation Frameworks (RAGAS and TruLens)

These concepts form the foundation of the AI Response Quality Evaluator Agent.

6.1 Large Language Model (LLM) Evaluation Techniques

Large Language Models (LLMs) are capable of generating natural language responses for a wide range of tasks, including question answering, text summarization, code generation, and conversational assistance. Although these models produce fluent and human-like text, their responses must be evaluated to ensure they are reliable and useful.

LLM evaluation is the process of measuring the quality of AI-generated responses using predefined metrics. Instead of checking only whether an answer is correct, modern evaluation considers multiple quality dimensions.

Common Evaluation Metrics

Accuracy

Accuracy measures whether the information provided by the AI is factually correct. A response is considered accurate when its statements match verified knowledge from trusted sources.

Example

Question: Who invented Java?

Correct Response: James Gosling developed Java at Sun Microsystems.

Incorrect Response: Dennis Ritchie invented Java.

The first response receives a high accuracy score, while the second is considered inaccurate.

Relevance

Relevance measures how closely the AI response answers the user's question. Even if the information is correct, it is not useful if it does not directly address the query.

Example

Question: What is Machine Learning?

Relevant Response: Machine Learning is a branch of Artificial Intelligence that enables systems to learn from data.

Irrelevant Response: Python is a programming language used in Machine Learning.

Although the second statement is true, it does not answer the question directly.

Completeness

Completeness evaluates whether the AI response includes all the important information needed to answer the question.

A complete response should:

- Cover the main concept
 - Include essential details
 - Avoid missing important points
-

Consistency

Consistency checks whether different parts of the response agree with each other. A response should not contain conflicting or contradictory statements.

Fluency

Fluency measures grammatical correctness, readability, and sentence structure. Responses should be easy to understand and professionally written.

Importance of LLM Evaluation

Effective evaluation helps developers:

- Improve response quality
 - Compare different language models
 - Detect weak responses
 - Increase user trust
 - Improve AI applications over time
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6.2 Hallucination Detection

One of the biggest challenges in Large Language Models is hallucination.

Hallucination occurs when an AI model generates information that appears convincing but is factually incorrect or unsupported by reliable evidence.

Unlike simple factual errors, hallucinated responses are often presented with high confidence, making them difficult for users to recognize.

Example

Question

Who discovered gravity?

Correct Answer

Sir Isaac Newton is credited with discovering the principles of gravity after observing a falling apple.

Hallucinated Response

Albert Einstein discovered gravity in 1925.

This response is incorrect because Einstein developed the theory of relativity, not the discovery of gravity.

Types of Hallucinations

1. Factual Hallucination

The AI generates incorrect facts.

Example:

"The capital of Australia is Sydney."

Correct answer: Canberra.

2. Fabricated Information

The AI invents names, dates, statistics, or references that do not exist.

Example:

"According to a 2024 NASA report, the Moon has active volcanoes."

No such report exists.

3. Unsupported Claims

The response contains statements that cannot be verified using trusted sources.

Hallucination Detection Process

The proposed system detects hallucinations by comparing the AI-generated response with trusted reference information retrieved through the RAG pipeline.

The evaluation process includes:

1. Retrieve trusted reference documents.
2. Compare response claims with retrieved information.
3. Identify unsupported statements.
4. Calculate a hallucination score.

5. Generate detailed feedback.

Benefits

- Reduces misinformation
- Improves AI reliability
- Increases user confidence
- Supports trustworthy AI systems

6.3 Retrieval-Augmented Generation (RAG)

Retrieval-Augmented Generation (RAG) is a technique that combines information retrieval with language models to improve response quality.

Instead of relying only on the knowledge stored inside the language model, RAG first retrieves relevant information from a trusted knowledge base and uses that information during evaluation.

This approach helps reduce hallucinations and improves factual accuracy.

Why RAG?

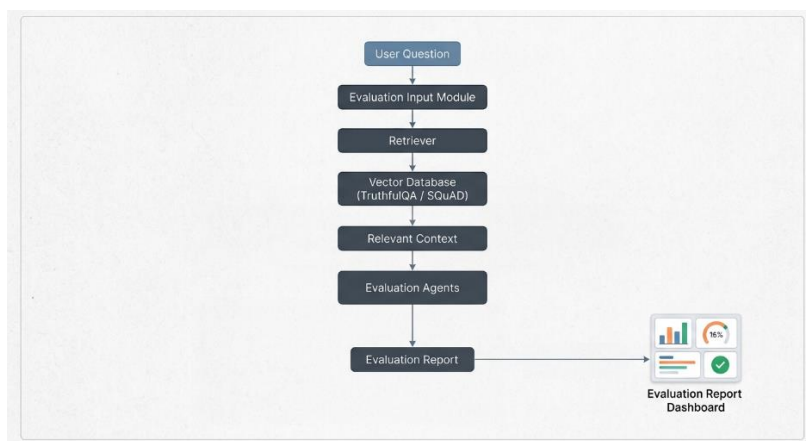
Traditional language models answer questions based on previously learned patterns. They cannot verify whether their response is supported by trusted evidence.

RAG solves this problem by retrieving relevant documents before evaluation.

Advantages include:

- Better factual accuracy
- Reduced hallucinations
- Support for updated knowledge
- Improved explainability
- Reliable evidence for evaluation

RAG Workflow



Components of RAG

Retriever

Searches the vector database and retrieves relevant documents.

Knowledge Base

Stores trusted information obtained from benchmark datasets.

Embedding Model

Converts text into numerical vectors for semantic search.

Vector Database

Stores embeddings and enables efficient similarity search.

Evaluation Agent

Uses retrieved information to verify AI-generated responses.

6.4 Existing Evaluation Frameworks

Several open-source frameworks are available for evaluating AI systems. The proposed project studies two widely used frameworks: **RAGAS** and **TruLens**.

RAGAS

RAGAS (Retrieval-Augmented Generation Assessment) is an evaluation framework specifically designed for RAG-based systems.

It measures how effectively retrieved information supports the generated response.

Main Metrics

- Faithfulness
- Answer Relevancy
- Context Precision
- Context Recall

Advantages

- Easy to integrate
- Designed for RAG systems
- Supports automated evaluation
- Provides quantitative scores

TruLens

TruLens is an evaluation framework for Large Language Model applications. It focuses on measuring response quality, groundedness, and correctness.

Main Metrics

- Correctness
- Groundedness
- Relevance
- Toxicity (optional)

Advantages

- Flexible evaluation pipeline
- Easy visualization
- Detailed feedback
- Suitable for production systems

Comparison of Evaluation Frameworks

Feature	RAGAS	TruLens
Primary Purpose	Evaluate RAG Systems	Evaluate LLM Applications
Accuracy Evaluation	✓	✓
Relevance Evaluation	✓	✓
Groundedness	✓	✓
Hallucination Detection	Indirect	Strong Support
Dashboard Support	Limited	Excellent
Best Use Case	RAG Evaluation	Production Monitoring

7. Proposed System Design

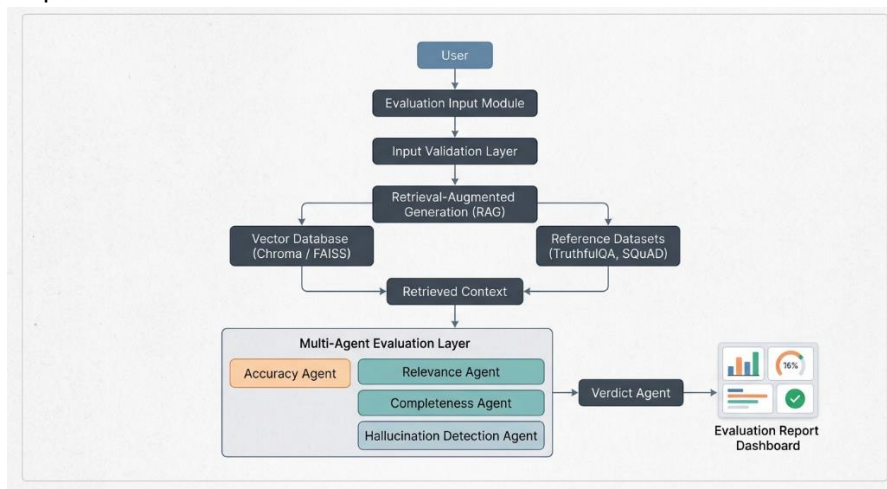
The AI Response Quality Evaluator Agent is designed as a modular, multi-agent system that evaluates the quality of AI-generated responses. The system combines Retrieval-Augmented Generation (RAG) with specialized evaluation agents to provide an accurate and explainable assessment. The design follows a layered architecture where each module has a specific responsibility. This modular approach improves scalability, maintainability, and future extensibility.

The overall evaluation process consists of the following stages:

1. Accept user input (question and AI response).
2. Retrieve relevant reference information from the knowledge base.
3. Evaluate the response using specialized agents.
4. Aggregate evaluation scores.
5. Generate the final evaluation report.

8. High-Level System Architecture

The proposed system consists of six major components that work together to evaluate AI-generated responses.



Architecture Explanation

The evaluation process begins when the user submits a question and its corresponding AI-generated response. The Evaluation Input Module validates the input and forwards it to the Retrieval-Augmented Generation (RAG) layer.

The RAG module retrieves the most relevant reference information from trusted datasets stored in the vector database. This retrieved context serves as the evidence for evaluating the AI response.

The response is then analyzed independently by four specialized evaluation agents. Each agent focuses on one quality dimension. Their scores and explanations are collected by the Verdict Agent, which calculates the overall quality score and generates the final evaluation report.

9. Agent Responsibilities

The system uses a multi-agent architecture where each agent has a clearly defined responsibility.

9.1 Evaluation Input Agent

Responsibility

The Evaluation Input Agent acts as the entry point of the system.

Functions

- Accept user question
- Accept AI-generated response
- Accept optional reference answer
- Validate input fields
- Forward valid input to the orchestration layer

9.2 Retrieval Agent

Responsibility

Retrieve trusted information required for evaluation.

Functions

- Generate embeddings
- Search vector database

- Retrieve Top-K relevant chunks
 - Pass retrieved context to evaluation agents
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9.3 Accuracy Agent

Responsibility

Measure factual correctness.

Functions

- Compare response with retrieved knowledge
 - Identify factual errors
 - Calculate accuracy score
 - Generate supporting explanation
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9.4 Relevance Agent

Responsibility

Measure how well the response answers the user's question.

Functions

- Analyze semantic similarity
 - Detect irrelevant information
 - Generate relevance score
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9.5 Completeness Agent

Responsibility

Determine whether the response covers all important information.

Functions

- Compare response with reference answer
 - Detect missing concepts
 - Generate completeness score
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9.6 Hallucination Detection Agent

Responsibility

Identify unsupported or fabricated information.

Functions

- Compare claims against retrieved evidence
 - Detect fabricated facts
 - Detect unsupported statements
 - Generate hallucination score
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9.7 Verdict Agent

Responsibility

Produce the final evaluation report.

Functions

- Collect scores from all agents
- Apply weighted scoring

- Calculate overall quality score
- Generate evaluation summary
- Produce user feedback

10. Scoring Dimensions

The proposed evaluation system measures AI responses across four primary quality dimensions.

Metric	Description	Weight (%)
Accuracy	Measures factual correctness	35
Relevance	Measures how well the response answers the question	25
Completeness	Measures coverage of important information	20
Hallucination Detection	Measures unsupported or fabricated claims	20

Overall Score Formula

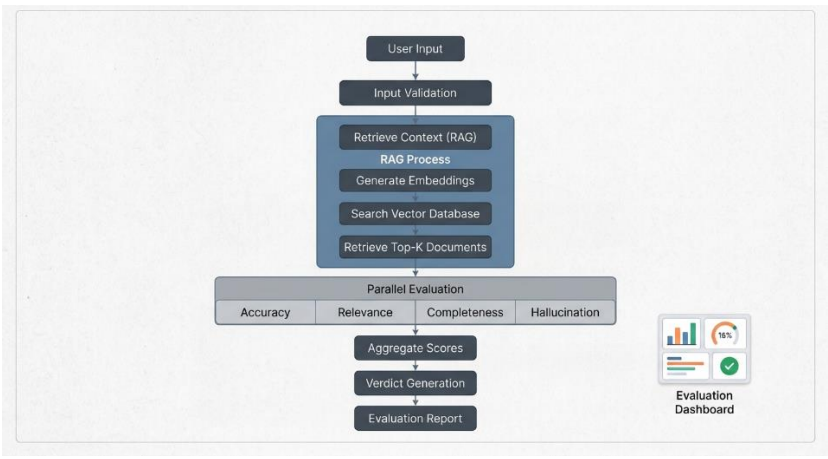
Overall Score =
(Accuracy × 0.35)
+ (Relevance × 0.25)
+ (Completeness × 0.20)
+ (Hallucination × 0.20)

This weighted scoring approach gives higher importance to factual correctness while still considering the relevance, completeness, and reliability of the response.

11. Orchestration Flow

The orchestration layer coordinates communication between all system components.

Workflow



Advantages of the Orchestration Layer

- Coordinates multiple evaluation agents
- Enables parallel execution
- Reduces evaluation time

- Improves scalability
 - Simplifies future extensions
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12. Data Models

The system uses four primary data models.

12.1 Submission Model

Field	Type	Description
Submission ID	UUID	Unique request identifier
Question	String	User question
AI Response	Text	Response generated by the LLM
Reference Answer	Text (Optional)	Trusted answer
Source Document	File / Text	Optional supporting document
Created At	Timestamp	Submission time

12.2 Judge Score Model

Field	Type	Description
Agent Name	String	Evaluation agent name
Score	Float	Score between 0 and 5
Confidence	Float	Confidence score
Rationale	Text	Explanation for assigned score
Evidence	JSON	Supporting evidence

12.3 Evaluation Result Model

Field	Type	Description
Overall Score	Float	Final quality score
Verdict	String	Pass / Needs Review / Fail
Dimension Scores	JSON	Individual scores

Field	Type	Description
Summary	Text	Overall evaluation summary
Evaluated At	Timestamp	Evaluation completion time

12.4 Knowledge Chunk Model

Field	Type	Description
Chunk ID	UUID	Unique chunk identifier
Source Dataset	String	TruthfulQA or SQuAD
Chunk Text	Text	Retrieved content
Embedding	Vector	Semantic representation
Metadata	JSON	Additional chunk information

13. Evaluation Input Module

13.1 Overview

The Evaluation Input Module serves as the entry point of the AI Response Quality Evaluator Agent. It provides a simple and user-friendly interface for submitting evaluation requests. The module collects the necessary inputs, validates them, and forwards the data to the orchestration layer for processing.

This module is designed to support both manual testing by users and future integration with APIs or other AI applications.

13.2 Input Fields

The system accepts the following inputs:

Field	Description	Required
Question	User's original question	Yes
AI Response	Response generated by the LLM	Yes
Reference Answer	Trusted answer for comparison	Optional
Source Document	Supporting document (PDF/Text)	Optional

13.3 Input Validation

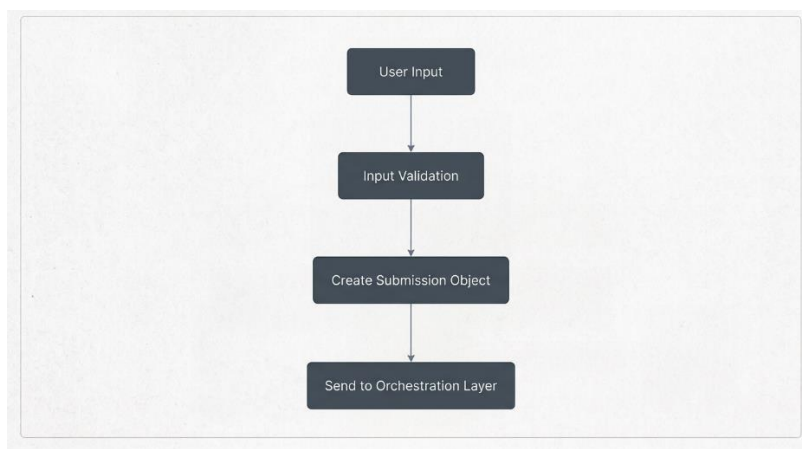
Before evaluation begins, the system validates all user inputs.

Validation includes:

- Ensuring the question is not empty.
- Ensuring the AI response is provided.
- Checking the format of uploaded documents.
- Limiting maximum input length.
- Removing unnecessary whitespace and invalid characters.

If validation fails, the system displays an appropriate error message to the user.

13.4 Functional Workflow



14. Reference Knowledge Base

14.1 Overview

The Reference Knowledge Base stores trusted information used to evaluate AI-generated responses. Instead of relying solely on the language model's internal knowledge, the evaluator retrieves relevant information from verified datasets.

This approach ensures that response evaluation is evidence-based and helps reduce hallucinations.

14.2 Datasets Used

TruthfulQA

TruthfulQA is a benchmark dataset designed to evaluate whether language models produce truthful answers instead of common misconceptions.

It contains hundreds of carefully designed questions covering topics such as science, history, health, politics, and everyday knowledge.

SQuAD (Stanford Question Answering Dataset)

SQuAD is a widely used benchmark dataset consisting of questions and answers extracted from Wikipedia articles.

It provides reliable reference answers for evaluating question-answering systems.

14.3 Knowledge Base Pipeline



14.4 Chunking Strategy

Large documents are divided into smaller sections called chunks before embedding.

Benefits:

- Faster retrieval
 - Better semantic matching
 - Improved search accuracy
 - Reduced memory usage
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14.5 Embedding Generation

Each text chunk is converted into a numerical vector using a sentence embedding model.

Embedding models capture the semantic meaning of text, enabling efficient similarity search in the vector database.

Example models include:

- Sentence Transformers
 - BAAI BGE Embeddings
 - OpenAI Embeddings
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14.6 Vector Database

The vector database stores embeddings and performs semantic similarity search.

Advantages:

- Fast retrieval
 - Scalable architecture
 - Efficient indexing
 - Supports millions of documents
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15. Technology Stack

The following technologies are proposed for implementing the AI Response Quality Evaluator Agent.

Component	Technology	Purpose
Programming Language	Python	Backend development
Frontend	HTML, CSS, JavaScript	User Interface
Backend Framework	FastAPI	REST API development
Agent Framework	LangGraph	Multi-agent orchestration
LLM Framework	LangChain	AI workflow integration
Embedding Model	Sentence Transformers	Generate vector embeddings
Vector Database	ChromaDB / FAISS	Store embeddings and retrieve context
Dataset	TruthfulQA, SQuAD	Trusted knowledge source
Version Control	Git & GitHub	Source code management
IDE	Visual Studio Code	Development environment

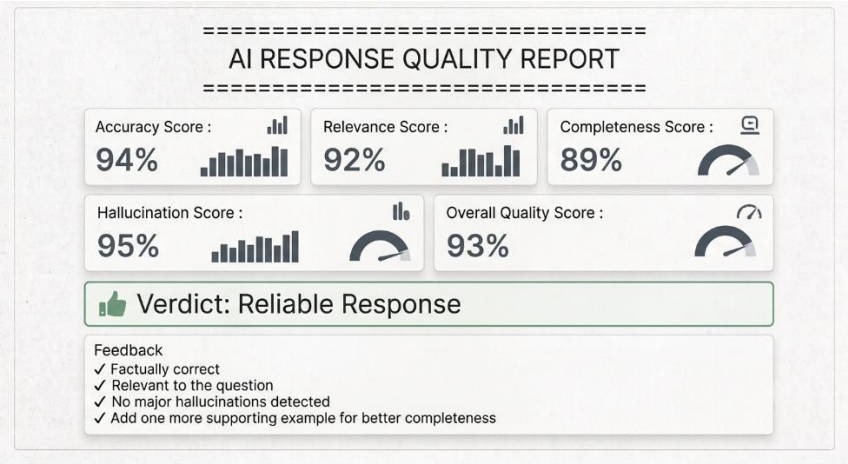
Why These Technologies?

- Python provides extensive AI and machine learning libraries.
 - FastAPI enables high-performance API development.
 - LangChain simplifies LLM integration.
 - LangGraph supports modular multi-agent workflows.
 - ChromaDB/FAISS allows efficient semantic search.
 - Sentence Transformers generate high-quality embeddings.
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16. Expected Output

After evaluation, the system generates a detailed report.

Sample Output



17. Future Enhancements

The current design establishes the foundation for evaluating AI-generated responses. Future versions of the project can include the following improvements:

- Support for multiple Large Language Models (ChatGPT, Gemini, Claude, Llama).
- Real-time evaluation through REST APIs.
- Interactive web dashboard with charts and analytics.
- Support for multilingual response evaluation.
- Integration with enterprise knowledge bases.
- Custom evaluation metrics for healthcare, finance, and legal domains.
- Automatic report generation in PDF format.
- User authentication and evaluation history tracking.

18. Conclusion

The AI Response Quality Evaluator Agent provides a systematic approach for evaluating the quality of AI-generated responses. By combining Retrieval-Augmented Generation (RAG) with a modular multi-agent architecture, the proposed system evaluates responses across multiple quality dimensions, including accuracy, relevance, completeness, and hallucination detection.

The modular design improves scalability, maintainability, and flexibility, allowing future integration of additional evaluation metrics and language models. The use of trusted datasets and vector-based retrieval enhances factual verification and reduces the risk of hallucinated responses.

This design document establishes a strong foundation for implementing a reliable and explainable AI evaluation system in future development phases.

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